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Recommendation by the Union of European Phoniatricians (UEP): Standardizing Voice Area Measurement/Phonetography

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Operational Definition

In phonetography the laryngeal possibilities with respect to fundamental frequency and sound intensity are registered. The frequency span on the horizontal axis of a phonetogram comprises the highest and the lowest frequencies at which the vocal folds are able to vibrate. On the vertical axis the maximum and minimum intensities of sound producible under controlled conditions of vowel production and mouth opening are registered at a certain distance from the mouth.

Application

The procedure may be used as part of an assessment of the normal voice [1]. It is useful for testing singing voice potentialities [2]. It may also be used as a diagnostic aid in cases of vocal disturbances and in the assessment of the results of therapeutic treatment [1].

Instrumental Equipment

Minimum instrumental equipment can be a simple sound level measuring device and a tone generator (for instance: keyboard musical instrument or electronic tone generator). The A-weighting network is advised (dB(A)), combined with slow meter damping as specified by the International Electrotechnical Commission [3]. A variation of the equipment consists of a sound level meter developed and built in Groningen, which switches automatically between two overlapping areas of 50 dB (35–85 and 75–125 dB). This makes readjustment of the sensitivity range of the sound level meter during (short) phonations unnecessary. In principle it is sufficient to sing the single tones to the test person. The subject has to sustain a laryngeal tone at the same fundamental frequency as loud and as soft as possible.

Maximum instrumental equipment nowadays consists of an X-Y writer, of which the X positioning may be done by using a voltage proportional to the frequency (log scale). This voltage may be taken from an electronic musical instrument or tone generator.

Positioning of the X axis of the writer by measuring the fundamental frequency directly from the acoustic signal demands a rather great effort and cannot be done easily with simple instruments. The Y positioning is obtained by measuring the sound pressure level with a linear decibel scale.

Computer-aided registration, including the evaluation of special features of tonal

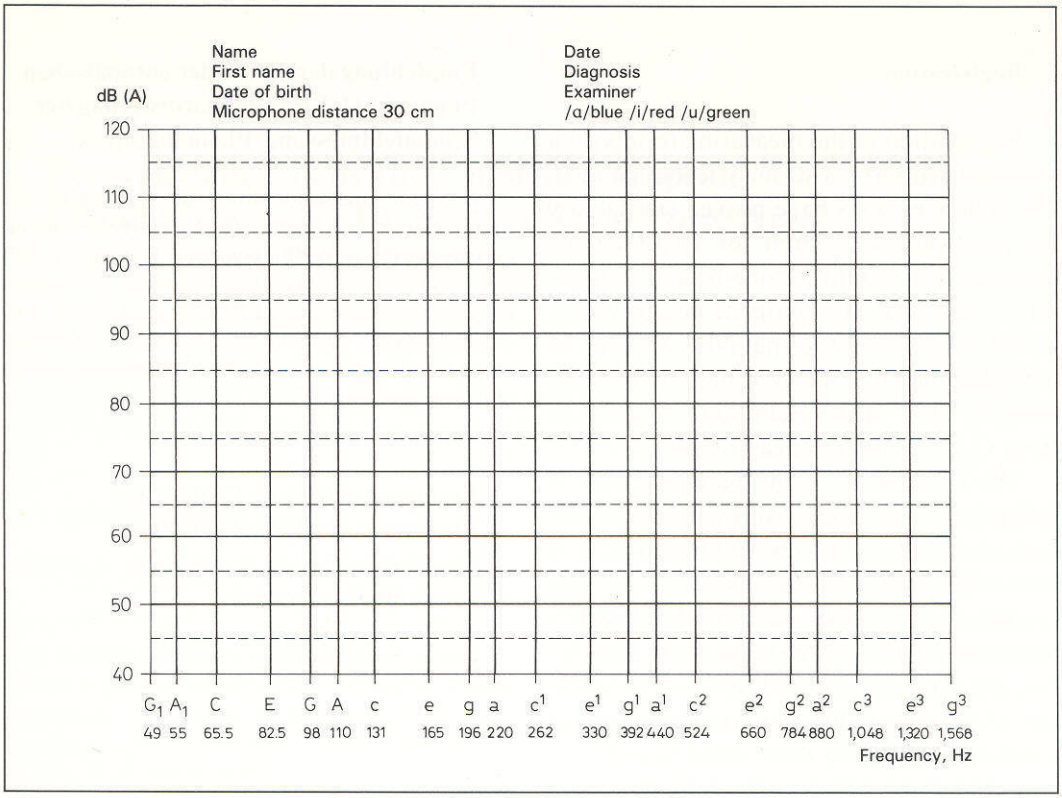

Fig. 1. Proposed dimensions of a standardized form for phonetography.

quality, has been designed and will be tested. In testing singers it is advisable to measure simultaneously the amplitude of the singer's formant at maximum sung intensity with a filter system [2]. The characteristics of this filter system are determined by the aim of the investigations. The microphone ought to have omnidirectional characteristics and should not be rigidly connected to the measuring apparatus.

Microphone Distance and Room Acoustics

At the recommended distance of 30 cm the microphone is capable of measuring the sound pressure level of the voice, even in low-intensity voice production, in spite of the commonly present surrounding noise, which then may not exceed 40 dB (A). In using this microphone distance, slight deviations of this distance are allowed in rooms with 'normal room acoustics' and rooms with strong reverberation.

Audiometer cabines or strongly damped experimental rooms may influence the auditive self-control of the subject's own voice; moreover, the microphone distance then is more critical.

Using an experimental room with 'living room acoustics' seems to be optimal and least critical with respect to the microphone distance.

Registration

Registration of the measuring results on a standardized form is strongly recommended. The following sizes have proved satisfactory for practical use: 15 mm for 10 dB on the ordinate and 36 mm for one-octave distance on the abscissa. The complete design, as depicted in figure 1 for format DIN A5, may be extended for DIN A4 by using a tone range from C_1 to c^5 in the Helmholtz notation or from C_1 to C_7 in American notation.

The registration of the phonetograms of different vowels appeared, especially in the case of singers, to be very informative. It is then sufficient to test the vowels /ɑ/, /u/ and /i/. We suggest in these cases a colored differentiation for the vowel-voice area: blue for /ɑ/, green for /u/, and red for /i/. The intensity of the singer's formant may be indicated in the same colors by interrupted lines [2]. In case of black and white presentations we suggest to use □ for /ɑ/, ○ for /u/, and △ for /i/.

Terminology

Until now the term 'Phonetogramm' has been used, but lately 'Stimmfeld' seems to prevail in German-speaking countries. In French originally the term 'courbe vocale' has been used. In English various terms have been published: phonetogram, voice profile, voice field, voice area, and F_o-SPL profile. In Dutch 'Stemgebied' seems possible, but the term 'Fonetogram' has become the most commonly used term. A definite terminological recommendation is lacking, however, the actually used term ought to take into account the fact that the results in a diagram represent a limited section of the frequency-sound level plane.

Empfehlung der Union der europäischen Phoniater (UEP): Standardisierung der Stimmfeldmessung/Phonetographie

Es wurde ein Standardisierungsvorschlag für die Stimmfeldmessung/Phonetographie erarbeitet und innerhalb der Union der Europäischen Phoniater (UEP) diskutiert and akzeptiert. Die vorgelegte Publikation gilt als Empfehlung der UEP und berücksichtigt Untersuchungsgegenstand, Geräteausstattung, Raumakustik, Mikrophonabstand, Registrierungsmöglichkeiten und Terminologie.

Recommandation de l'Union des Phoniatres Européens (l'UPE): Standardisation de la mesure de la courbe vocale/phonétographie

Une proposition relative à la standardisation de la mesure de la courbe vocale/phonétographie a été discutée et acceptée dans le cadre d'une réunion de l'Union des Phoniatres Européens (UPE). La présente publication est une recommandation de la part de l'UPE à propos des sujets suivants: l'objet de recherche, l'instrumentation minimale, l'acoustique des salles, la distance du microphone, les possibilités d'enregistrement graphique et la terminologie.

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